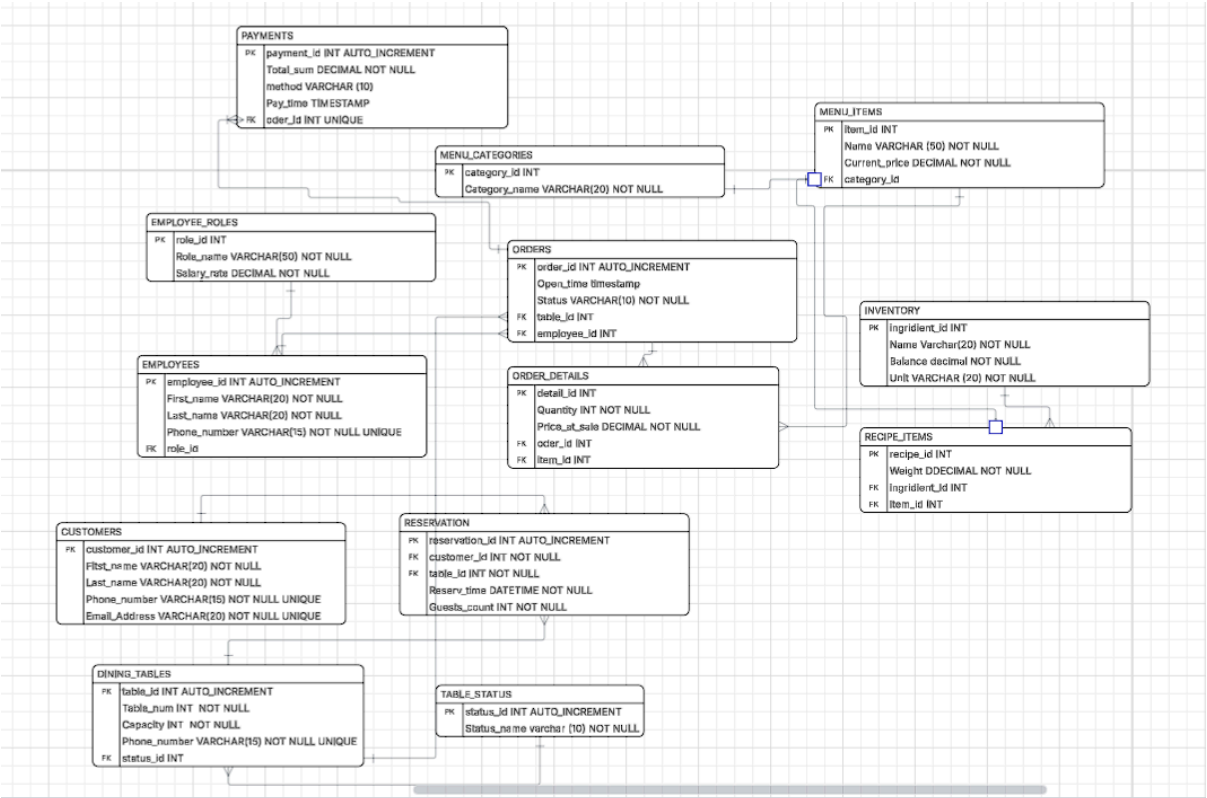
1.1 Logical Scheme



1.2 Objects

Table – Customers

Field Name	Field Description	Data Type
customer_id	Unique identifier of the customer	INT PRIMARY KEY AUTO_INCREMENT
First_name	Customer's name	VARCHAR(20) NOT NULL
Last_name	Customer's surname	VARCHAR(20) NOT NULL
Phone_number	Customer's phone number	VARCHAR(10) NOT NULL
Email_Address	Email address of Customer	VARCHAR(36) NOT NULL

Comments on table relationships: 1:N relationship with the Reservations table.

Table – Reservation

Field name	Field Description	Data Type
reserv_id	Unique identifier for the booking	INT PK
reserv_time	Date and time of the scheduled visit	DATETIME NOT NULL
guests_count	Number of expected guests	INT NOT NULL
customer_id	Reference to the customer making the booking	INT FK
table_id	Reference to the assigned table	INT FK

Comments on table relationships: 1:N relationship with DiningTables.

Table – Table_Status

Field Name	Field Description	Data Type
status_id	Unique identifier for the status	INT PK
status_name	Name of state (Available, Occupied, Reserved	VARCHAR (10) NOT NULL

Comments on table relationships: Linked to TableStatus (FK) and Orders.

Table – Dining_Tables

Filed Name	Field Description	Data Type
table_id	Unique identifier for the table	INT PK
table_number	The number assigned to the table in the hall	INT NOT NULL
Capacity	Maximum seating capacity of the table	INT NOT NULL

status_id	Reference to current table state	INT FK

Comments on table relationships: Junction between `Customers` and `DiningTables`.

Table – Empolyee_roles

Field Name	Field Description	Data Type
role_id	Unique identifier for the role	INT PK
role_name	Job title (e.g., Waiter, Chef, Manager)	VARCHAR (30) NOT NULL
Salary_rate	Standard hourly pay rate for this position	DECIMAL (10, 2)NOT NULL

Comments on table relationships: 1:N relationship with Employees.

Tablee – Employees

Field Name	Field Description	Data Type
employee_id	Unique staff identification number	INT PRIMARY KEY AUTO_INCREMENT
First_name	Employee's name	VARCHAR(20) NOT NULL
Last_name	Employee's surname	VARCHAR(20) NOT NULL
Phone_number	Employee's phone number	VARCHAR(10) NOT NULL
role_id	Reference to the job position	FK INT

Comments on table relationships: Linked to EmployeeRoles (FK) and Orders.

Table – Orders

Field Name	Filed Description	Data Type
order_id	Unique order number	INT PK
table_id	Reference to the table where order is placed	INT FK
employee_id	Reference to staff member serving the order	INT FK
open_time	Time when the order was opened	TIMESTAMP NOT NULL
status	Current order status (e.g., Active, Served)	VARCHAR (20)

Comments on table relationships: Central hub connecting tables, staff, and payments.

Table – Menu_Categories

Field Name	Field Description	Data Type
category_id	Unique identifier for the category	INT PK
category_name	Category name (e.g., Appetizers, Drinks)	VARCHAR(50) NOT NULL

Comments on table relationships: 1:N relationship with MenuItems.

Table – Menu_Items

Field Name	Field Description	Data Type

item_id	Unique identifier for the menu item	INT PK
name	Name of the dish or drink	VARCHAR(100) NOT NULL
current_price	The standard selling price of the item	DECIMAL(10,2) NOT NULL
category_id	Reference to the menu category	INT FK

Comments on table relationships: Linked to categories and M:N bridge tables.

Table – Order_Details

Field Name	Field Description	Data Type
detail_id	Unique line-item identifier	INT PK
order_id	Reference to the parent order	INT FK
item_id	Reference to the ordered menu item	INT FK
quantity	Number of portions ordered	INT NOT NULL

price_at_sale	The price recorded at the time of purchase	DECIMAL(10,2) NOT NULL

Comments on table relationships: Implements Many-to-Many between Orders and

Table – Payments

Field Name	Field Description	Data Type
payment_id	Unique transaction identifier	INT PK
order_id	Reference to the order	INT FK UNIQUE
total_sum	Final total amount paid	DECIMAL(10,2) NOT NULL
method	Payment method (e.g., Card, Cash)	VARCHAR(20)
pay_time	Time the payment was processed	TIMESTAMP

Comments on table relationships: 1:1 relationship with Orders.

Table – Inventory

Field Name	Field Description	Data Type
ingredient_id	Unique identifier for the ingredient	INT PK
name	Name of the raw product (e.g., Flour, Milk)	VARCHAR(100) NOT NULL
balance	Current physical quantity in stock	DECIMAL(12,3) NOT NULL
unit	Unit of measure (e.g., kg, l)	VARCHAR(10)

Comments on table relationships: Linked to RecipeItems for stock deduction.

Table – Recipt_Items

Field Name	Field Description	Data Type
recipe_id	Unique identifier for the recipe entry	INT PK
item_id	Reference to the dish in the menu	INT FK
ingredient_id	Reference to the raw ingredient	INT FK

weight	Amount of ingredient required for one portion	DECIMAL(12,3) NOT NULL

Comments on table relationships: Implements Many-to-Many between MenuItems and Inventory.

Example of “Customers” Table :

customer_id	First_Name	Last_Name	Phone_number	Emaill_address
5001	Azamat	Lopin	+50564896	A.L00p1n@gmail.com
5002	Zhalgas	Saakashvili	+58950578	saakashv1l1@gmail.com